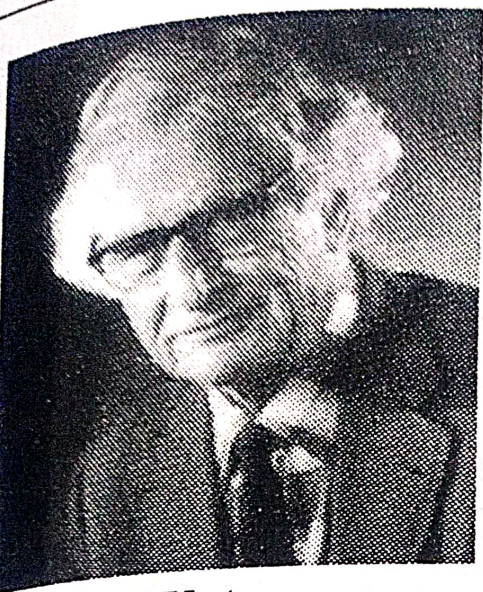


Chapter 2...

Chi-Square (χ^2) Distribution



Yates

Born on May 12 1902 in Manchester England he was the eldest of five children. He began his mathematic education at a private school where he was encouraged and taught by an advanced mathematics teacher. Yates received a scholarship to Clifton College and was later awarded another scholarship at St. Johns College in Cambridge. He taught mathematics for a short period of time but wanted to use mathematics to help others instead of just

teaching it. A few years later he went to Africa where he was part of the old coast survey. A few years later he moved back and was appointed assistant statistician at Rothamsted Experimental Station. When a friend, A. Fisher became chairman at the University College London in 1933, Yates was appointed the Head of Statistics at Rothamsted. He held that position until 1968 when he retired. During the time he was working, he worked often with Fisher on experimental design, block designs, theory of variance and operational research.

Contents ...

- 0 Introduction
- 1 Derivation of P.D.F. of χ^2 with n Degrees of Freedom
- 2 Moments of Chi-square Distribution
- 3 Recurrence Relation among the Central Moments
- 4 Additive Property
- 5 Mode of the Chi-square Distribution
- 6 Use of χ^2 Tables for Calculation of Probabilities
- 7 Limiting behaviour of χ^2 as $n \rightarrow \infty$. [Normal Approximation]
- 8 Results Based on Bivariate Transformation
- 9 Probability Distribution of $\bar{X}$ and S^2

Key Words :

Gamma distribution, chi-square distribution, degrees of freedom (d.f.), normal approximation.

Objectives :

- (1) To understand chi-square distribution as a particular case of gamma distribution.
- (2) To study the relationship between chi-square distribution and other distributions. Also limiting behaviour of chi-square distribution as $n \rightarrow \infty$ (normal approximation).
- (3) Computation of probabilities using χ^2 tables and also using MS-EXCEL, R-software.
- (4) To study the properties of χ^2 distribution.

2.0 Introduction

The gamma distribution is related with the theory of independent normal variates in natural way. Because we have shown that if $X_1, X_2, \dots, X_i, \dots, X_n$ are n independent standard normal variates then the distribution of $\sum_{i=1}^n X_i^2$ has $G\left(\frac{1}{2}, \frac{n}{2}\right)$ distribution. This particular case of gamma distribution is called as *chi-square* distribution with n degrees of freedom (d.f.). The random variable involved in this distribution is denoted as χ_n^2 . The degrees of freedom (d.f.) represents the number of independent variables used in the construction of χ^2 . Thus χ_n^2 variate can be looked-upon as the sum of squares of n independent standard normal variates.

Tests based on chi-square distribution like chi-square test of goodness of fit, chi-square test of independence of attributes etc. are widely used. It also plays an important role in statistical inference.

2.1 Derivation of P.D.F. of χ^2 with n Degrees of Freedom (using M.G.F.)

Let $X_1, X_2, \dots, X_i, \dots, X_n$ be i.i.d. $N(0, 1)$ variates. Then m.g.f. of χ^2 is given by,

$M_{\chi^2}(t) = \int_{-\infty}^{\infty} e^{tx^2} f(x) dx$ where, $f(x)$ is p.d.f. of standard normal distribution.

$$\begin{aligned} \therefore M_{\chi^2}(t) &= \int_{-\infty}^{\infty} e^{tx^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-\left(\frac{1}{2}-t\right)x^2} dx \end{aligned}$$

[$\because$ Integrand is even function of x , Also $a = \frac{1}{2} - t, b = 1$]

$$= \frac{2}{\sqrt{2\pi}} \cdot \frac{1}{2} \frac{\Gamma\left(\frac{1}{2}\right)}{\left(\frac{1}{2}-t\right)^{1/2}} ; \frac{1}{2} - t > 0$$

$$\begin{aligned} &= \frac{2}{\sqrt{2\pi}} \cdot \frac{1}{2} \frac{\Gamma\left(\frac{1}{2}\right)}{(1-2t)^{1/2}} 2^{1/2} = \left(\frac{1}{1-2t}\right)^{1/2} \\ &= (1-2t)^{-1/2} \end{aligned}$$

$$\therefore M_{\chi^2}(t) = \left(1 - \frac{t}{1/2}\right)^{-1/2} ; t < \frac{1}{2}$$

$$\therefore \text{If } X_i \text{ is } N(0, 1) \text{ variate } M_i^2(t) = \left(1 - \frac{t}{1/2}\right)^{-1/2} ; t < \frac{1}{2}$$

Suppose, $Y = \sum_{i=1}^n X_i^2$. Then Y has χ^2 distribution with n d.f. (χ_n^2).

The m.g.f. of Y is as follows :

$$\begin{aligned} M_Y(t) &= M_{\sum_{i=1}^n X_i^2}(t) \\ &= M_{X_1^2}(t) \cdot M_{X_2^2}(t) \dots M_{X_i^2}(t) \dots M_{X_n^2}(t) \end{aligned}$$

$$= \left(1 - \frac{1}{1/2}\right)^{\frac{n}{2}}$$

[$\because X_i$'s are independent]

It is m.g.f. of $G\left(\frac{1}{2}, \frac{n}{2}\right)$. Thus $\sum_{i=1}^n X_i^2 = Y$ has $G\left(\frac{1}{2}, \frac{n}{2}\right)$. Hence p.d.f. Y is given as.

$$f(y) = \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} e^{-\frac{y}{2}} y^{\frac{n}{2}-1} ; y \geq 0$$

$$= 0;$$

$$\text{OR } f(y) = \frac{1}{(\sqrt{2})^n \Gamma\left(\frac{n}{2}\right)} e^{-\frac{y}{2}} y^{\frac{n}{2}-1} ; y \geq 0$$

otherwise

... (1)

Thus a continuous variable Y is said to follow chi-square distribution with n d.f. if its p.d.f. is given by relation (1).

Another method :

Let $X_1, X_2, \dots, X_i \dots X_n$ be i.i.d. $N(0, 1)$ variates.

Suppose $Y_i = X_i^2$ we shall obtain distribution function of Y . It is given by

$$G_Y(y) = P[Y \leq y] = P[X_i^2 \leq y]$$

$$= P(-\sqrt{y} \leq X_i \leq \sqrt{y})$$

$$= F(\sqrt{y}) - F(-\sqrt{y}) \quad [F(\cdot) \text{ is distribution function of } X_i]$$

In order to obtain p.d.f. of y , we take derivatives of both sides w.r.t. y .

$$g(y) = \frac{dG(y)}{dy} = \frac{d}{dy} [F(\sqrt{y}) - f(-\sqrt{y})]$$

$$= \frac{f(\sqrt{y})}{2\sqrt{y}} - \frac{f(-\sqrt{y})}{2\sqrt{y}} (-1)$$

$$= \frac{f(\sqrt{y}) + f(-\sqrt{y})}{2\sqrt{y}} \quad [\because X \rightarrow N(0, 1), f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}]$$

$$= \frac{\frac{1}{\sqrt{2\pi}} e^{-y/2} + \frac{1}{\sqrt{2\pi}} e^{-y/2}}{2\sqrt{y}}$$

$$\therefore g(y) = \frac{\frac{1}{\sqrt{2\pi}} e^{-y/2}}{\sqrt{y}} ; y > 0$$

$$= \frac{\left(\frac{1}{2}\right)^{1/2}}{\Gamma\left(\frac{1}{2}\right)} e^{-y/2} y^{\frac{1}{2}-1} ; y > 0 \quad [\because \Gamma\frac{1}{2} = \sqrt{\pi}]$$

which p.d.f. of gamma distribution with parameters $\left(\frac{1}{2}, \frac{1}{2}\right)$. Thus X_i^2 follows $G\left(\frac{1}{2}, \frac{1}{2}\right)$ distribution $i = 1, 2 \dots n$.

$\therefore$ Each X_i^2 follows $G\left(\frac{1}{2}, \frac{1}{2}\right)$ distribution. Using additive property

of gamma variates.

$T = \sum_{i=1}^n X_i^2$ follows $G\left(\frac{1}{2}, \frac{n}{2}\right)$ distribution. The p.d.f. of T which has χ_n^2 probability distribution is given by

$$f(t) = \frac{1}{(\sqrt{2})^n \Gamma\left(\frac{n}{2}\right)} e^{-t/2} t^{\frac{n}{2}-1} ; t \geq 0$$

Remarks :

1. The probability density curves for $n = 1, 2, 3, 4, 5$, and 6 are as shown in figure 1.1.

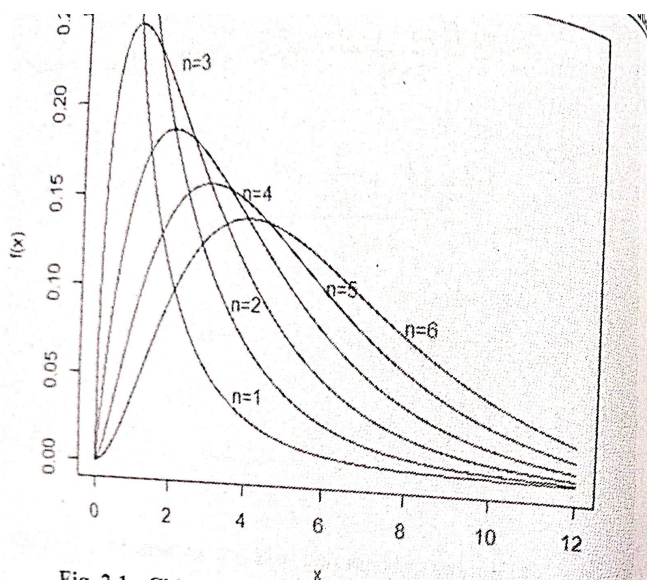


Fig. 2.1 : Chi-square probability density function

The probability density curve of χ^2 distribution has longer tail on right hand side. Thus the probability density curve is positively skew. It is also clear from Fig. 2.1 that mode of chi-square distribution exists only when $n > 2$.

2. Some practical situations in which χ^2 variate occurs can also be stated. For instance, suppose X represents the error in measuring radius of a circle and $X \rightarrow N(0, 1)$. Then error in measuring area of the circle = $A = \pi X^2$

$\therefore Y = \frac{A}{\pi} = X^2$ follows chi-square distribution with 1 d.f.

2.2 Moments of Chi-Square Distribution

If $Y \rightarrow \chi_n^2$, then r^{th} raw moment of Y is given by,

$$\mu'_r = E(Y^r) = \int_0^\infty y^r f(y) dy = \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma\left(\frac{n}{2}\right)} \int_0^\infty y^r e^{-\frac{y}{2}} y^{\frac{n}{2}-1} dy$$

$$= \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma\left(\frac{n}{2}\right)} \int_0^\infty e^{-\frac{y}{2}} y^{\frac{n}{2}+r-1} dy = \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma\left(\frac{n}{2}\right)} \cdot \frac{\Gamma\left(\frac{n}{2}+r\right)}{\left(\frac{1}{2}\right)^{\frac{n}{2}+r}}$$

$$\therefore \mu'_r = \frac{\Gamma\left(\frac{n}{2}+r\right)}{\Gamma\left(\frac{n}{2}\right)} \cdot 2^r \quad \dots (i)$$

$$= 2^r \left(\frac{n}{2}+r-1\right) \left(\frac{n}{2}+r-2\right) \dots \left(\frac{n}{2}+1\right) \left(\frac{n}{2}\right)$$

$$\therefore \mu'_r = n(n+2) \dots (n+2r-2); r = 1, 2, 3 \dots \quad \dots (ii)$$

putting $r = 1, 2, 3$ and 4 in equation (ii) we get first four raw moments as;

$$\mu'_1 = \text{Mean} = n$$

$$\mu'_2 = n(n+2)$$

$$\mu'_3 = n(n+2)(n+4)$$

$$\mu'_4 = n(n+2)(n+4)(n+6)$$

Remarks :

1. From above formulae we have

$$\begin{aligned} \text{Var}(Y) &= \mu'_2 - (\mu'_1)^2 \\ &= n(n+2) - n^2 = 2n \end{aligned}$$

It is convenient to use cumulant generating function (c.g.f.) to obtain central moments.

2. Recurrence relation between raw moments can be obtained from equation (i). Replacing r by $(r-1)$ in equation (i) we get,

$$\mu'_{r-1} = \frac{\Gamma\left(\frac{n}{2}+r-1\right)}{\Gamma\left(\frac{n}{2}\right)} \cdot 2^{r-1}$$

$$\frac{\mu'_r}{\mu'_{r-1}} = \left(\frac{n}{2} + r - 1\right) \cdot 2$$

$$\therefore \mu'_r = 2 \left(\frac{n}{2} + r - 1\right) \cdot \mu'_{r-1}; r = 1, 2, 3, \dots$$

For example : Putting $r = 1$, $\mu'_1 = n$, $r = 2$, $\mu'_2 = 2 \left(\frac{n}{2} + 1 - 1\right) \mu'_1 = (n+2)n$ etc.

M.G.F. and C. G. F. of χ^2 distribution

Let $Y \rightarrow \chi^2_n$. Then m.g.f. of Y is given by

$$M_Y(t) = E[e^{ty}]$$

$$= \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} \int_0^\infty e^{ty} e^{-\frac{y}{2}} y^{\frac{n}{2}-1} dy$$

$$= \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} \int_0^\infty e^{-\left(\frac{1}{2}-t\right)y} y^{\frac{n}{2}-1} dy, \quad \text{if } \left(\frac{1}{2}-t\right) > 0$$

$$= \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} \cdot \frac{\Gamma\left(\frac{n}{2}\right)}{(1-2t)^{n/2}} \cdot 2^{\frac{n}{2}}, \quad \left(\text{if } t < \frac{1}{2}\right)$$

$$M_Y(t) = (1-2t)^{-\frac{n}{2}},$$

In order to obtain raw moments, we expand R.H.S. of above equation in terms of powers of t .

$$\therefore M_Y(t) = 1 - \left(\frac{-n}{2}\right)(2t) + \frac{\left(\frac{-n}{2}\right)\left(\frac{-n}{2}-1\right)}{2!} 2^2 t^2 + \dots + \frac{(-1)^r \left(\frac{-n}{2}\right)\left(\frac{-n}{2}-1\right)\dots\left(\frac{-n}{2}-r+1\right)}{r!} 2^r t^r + \dots$$

$$\therefore \mu'_r = \text{coefficient of } \frac{t^r}{r!} = n(n+2) \dots (n+2r-2)$$

Note : Putting $r = 1, 2, 3, 4$ the above expression we get first four raw moments as $\mu'_1 = n$, $\mu'_2 = n(n+2)$, $\mu'_3 = (n+2)(n+4)$ and $\mu'_4 = n(n+2)(n+4)(n+6)$

The cumulant generating function is given by

$$K_Y(t) = \ln M_Y(t)$$

$$= -\frac{n}{2} \ln(1-2t)$$

$$= \frac{n}{2} \left\{ 2t + \frac{2^2 t^2}{2} + \frac{2^3 t^3}{3} + \frac{2^4 t^4}{4} + \dots + \frac{2^r t^r}{r} + \dots \right\}$$

$$\therefore \text{First cumulant} = k_1 = \text{Coefficient of } \frac{t^1}{1!} = n$$

$$\text{Second cumulant} = k_2 = \text{Coefficient of } \frac{t^2}{2!} = 2n$$

$$\text{Third cumulant} = k_3 = \text{Coefficient of } \frac{t^3}{3!} = 8n$$

$$\text{Fourth cumulant} = k_4 = \text{Coefficient of } \frac{t^4}{4!} = (3)2^4 n = 48n$$

$$\text{In general } r^{\text{th}} \text{ cumulant} = k_r = \text{coefficient of } \frac{t^r}{r!}$$

$$= n(r-1)! 2^{r-1}; \quad r = 1, 2, \dots$$

$$= (r-1)! n 2^{r-1}; \quad r = 1, 2, \dots$$

Remark : Then the first four central moments are obtained as follows :

$$\mu_1 = 0, \quad \mu_2 = k_2 = 2n, \quad \mu_3 = k_3 = 8n$$

$$\mu_4 = k_4 + 3k_2^2 = 48n + 12n^2$$

Coefficients of Skewness and Kurtosis

The coefficients of skewness β_1 and γ_1 are given by,

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(8n)^2}{(2n)^3} = \frac{8}{n};$$

$$\gamma_1 = \sqrt{\beta_1} = \sqrt{\frac{8}{n}}$$

Hence as $n \rightarrow \infty$, $\beta_1 \rightarrow 0$ and $\gamma_1 \rightarrow 0$. It means that when d.f. is very large, the distribution becomes approximately symmetric.

The coefficients of Kurtosis β_2 and γ_2 are,

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{48n + 12n^2}{4n^2} = 3 + \frac{12}{n}$$

$$\text{and } \gamma_2 = \beta_2 - 3 = \frac{12}{n}$$

Therefore as $n \rightarrow \infty$, $\beta_2 \rightarrow 3$ and $\gamma_2 \rightarrow 0$.

Thus distribution becomes mesokurtic for large d.f. (n).

2.3 Recurrence Relation Among the Central Moments

Let Y follow χ^2 distribution with n. d.f.

Then the m.g.f. is given by,

$$M_Y(t) = (1 - 2t)^{-\frac{n}{2}}$$

Hence the m.g.f. for obtaining central moments.

(Central m.g.f.) is given by, $M_{Y-E(Y)}(t) = M_{Y-n}(t) = e^{-nt} M_Y(t)$

$$M_{Y-n}(t) = e^{-nt} (1 - 2t)^{-\frac{n}{2}} \quad \therefore E(Y) = n$$

Taking logarithms of both sides

$$\ln M_{Y-n}(t) = -nt - \frac{n}{2} \ln(1 - 2t)$$

Differentiating both sides w.r.t. t we get

$$\begin{aligned} \frac{M'_{Y-n}(t)}{M_{Y-n}(t)} &= -n + \frac{n}{2} \times \frac{2}{1-2t} \\ &= \frac{2nt}{1-2t} \end{aligned}$$

$$\therefore (1-2t) M'_{Y-n}(t) = 2nt M_{Y-n}(t)$$

Differentiating both sides w.r.t. t, r times

$$(1-2t) M_{Y-n}^{r+1}(t) - 2r M_{Y-n}^r(t) + \dots = 2n$$

$$\left[t M_{Y-n}^r(t) + r M_{Y-n}^{r-1}(t) + \dots \right]$$

where,

$$M^r(t) = \frac{d^r M(t)}{d t^r}$$

= rth derivative of M(t) w.r.t. t.

Putting $t = 0$, we have

$$\mu_{r+1} - 2r \mu_r = 0 + 2n r \mu_{r-1}$$

$$\mu_{r+1} = 2r [n \mu_{r-1} + \mu_r] \text{ for } r = 1, 2, 3, \dots$$

Note : The above recurrence relation can be directly proved using integrals.

Deductions : $\mu_0 = 1, \mu_1 = 0$

$\therefore$ Putting $r = 1$ in above relation $\mu_2 = 2 [n \mu_0 + \mu_1] = 2n$

Putting $r = 2$ and $r = 3$ we obtain μ_3 and μ_4 as

$$\mu_3 = 8n, \quad \mu_4 = 48n + 12n^2$$

2.4 Additive Property

Statement : If Y_1 and Y_2 are independent χ^2 variates with n_1 and n_2 d.f. respectively, then $Y_1 + Y_2$ has also χ^2 distribution with $(n_1 + n_2)$ degrees of freedom.

Proof : $Y_1 \rightarrow \chi_{n_1}^2$

$\therefore$ The m.g.f. of $Y_1 = M_{Y_1}(t) = (1 - 2t)^{-n_1/2}$

Similarly, $Y_2 \rightarrow \chi_{n_2}^2$

$\therefore$ The m.g.f. of $Y_2 = M_{Y_2}(t) = (1 - 2t)^{-n_2/2}$

The m.g.f. of $Y_1 + Y_2$ is given by

$$M_{Y_1+Y_2}(t) = M_{Y_1}(t) \cdot M_{Y_2}(t) \quad (\because Y_1 \text{ and } Y_2 \text{ are independent})$$

$$= (1 - 2t)^{-n_1/2} (1 - 2t)^{-n_2/2}$$

$$= (1 - 2t)^{-\frac{n_1 + n_2}{2}}$$

which is m.g.f. of χ^2 distribution with $n_1 + n_2$ d.f.

Thus $Y_1 + Y_2 \rightarrow \chi_{n_1 + n_2}^2$

Remark : Generalisation of the above stated property holds good. Hence if $Y_1, Y_2, \dots, Y_k$ are independent χ^2 variates with $n_1, n_2, \dots, n_k$ d.f. respectively then $\sum_{i=1}^k Y_i$ follows χ^2 distribution with $\sum_{i=1}^k n_i$ d.f.

2.5 Mode of the Chi-square Distribution

The value of y which maximizes $f(y)$ is the mode of the distribution. Hence, mode will be the value such that $f'(y)_{y=M_0} = 0$ and $f''(y)_{y=M_0} < 0$.

Let Y be a χ^2 variate with n d.f.

Then the p.d.f. is given by,

$$f(y) = \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma\left(\frac{n}{2}\right)} e^{-y/2} y^{\frac{n}{2}-1}; y > 0$$

$$\therefore \log_e f(y) = \log_e c - \frac{y}{2} + \left(\frac{n-2}{2}\right) \log_e y$$

$$\text{where, } c = \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma\left(\frac{n}{2}\right)}$$

Differentiating both sides w.r.t. y we have

$$\frac{f'(y)}{f(y)} = 0 - \frac{1}{2} + \frac{n-2}{2y}$$

$$\therefore f'(y) = \frac{f(y)}{2} \left[\frac{n-2}{y} - 1 \right]$$

$$\therefore f'(y) = 0 \Rightarrow \frac{f(y)}{2} \left[\frac{n-2}{y} - 1 \right] = 0$$

$$y = n-2$$

.... (i)

$$f''(y) = \frac{f'(y)}{2} \left[\frac{n-2}{y} - 1 \right] + \frac{f(y)}{2} \left[-\frac{(n-2)}{y^2} \right]$$

$$[f''(y)]_{y=n-2} = 0 - \frac{f(y)(n-2)}{2y^2} < 0 \quad \text{for } n > 2 \quad \dots (ii)$$

From equations (i) and (ii) mode of the distribution is $n-2$.

Note : From Fig. 2.1, we can get the idea that mode of χ_n^2 exists for $n > 2$.

2.6 (A) Use of χ^2 Tables for Calculation of Probabilities

Let Y be a χ^2 variate with n d.f. We need to find the probabilities such as $P[Y \geq a]$, $P[Y \leq b]$, $P[c \leq y \leq d]$ where a, b, c, d are constants. The computations of these probabilities by evaluating the integral is a very difficult task. However, the computations of these probabilities is facilitated by using the tables giving values of χ^2 variates for various d.f. from 1 to 70. [Table of distribution of χ^2 from statistical table].

Suppose $\chi_{n, \alpha}^2$ represents the value of χ^2 variate such that

$$P[Y \geq \chi_{n, \alpha}^2] = \int_{\chi_{n, \alpha}^2}^{\infty} \frac{1}{2^n \Gamma\left(\frac{n}{2}\right)} y^{n/2-1} e^{-y/2} dy = \alpha$$

Then values of $\chi_{n, \alpha}^2$ are available in the table for $\alpha = 0.99, 0.98, 0.95, 0.90, 0.80, 0.70, 0.50, 0.30, 0.20, 0.10, 0.05, 0.2, 0.01$ and 0.001 . The area corresponding to this probability is shaded area as shown in Fig. 2.2.

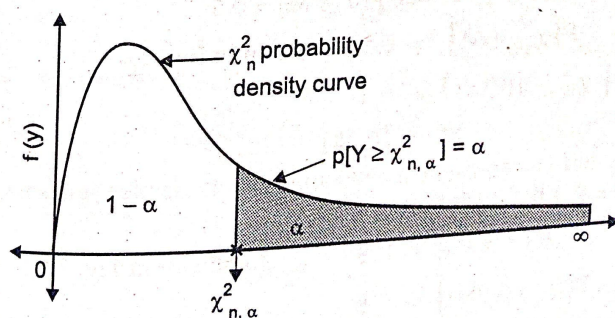


Illustration 1 : Let $Y = \chi^2_{11}$. Find (i) $P[Y > 14.631]$ (ii) $P[Y \leq 8.148]$ (iii) $P[4.575 < Y < 17.275]$.
 (iv) Median of distribution of Y , (v) Second and seventh deciles of distribution of Y .

Solution :

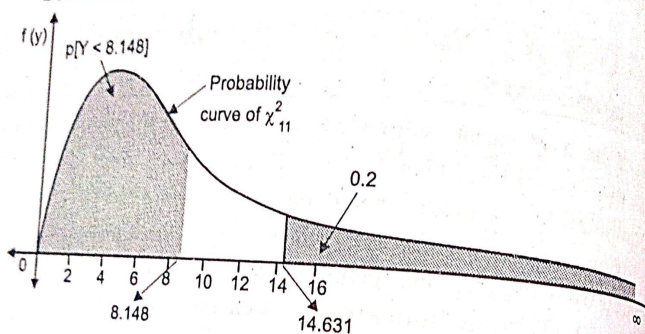


Fig. 2.3

(i) $P[Y > 14.631] = 0.2$

[From statistical table]

(ii) $P[Y \leq 8.148] = P[\chi^2_{11} \leq 8.148] = 1 - P[\chi^2_{11} > 8.148]$
 $= 1 - 0.7 = 0.3$

(iii) $P[4.575 < Y < 17.275]$

$= P[4.575 < \chi^2_{11} < 17.275]$

$= P[\chi^2_{11} \geq 4.575] - P[\chi^2_{11} \geq 17.275]$

$= 0.95 - 0.1 = 0.85$

(iv) Suppose median = M , then $P[Y \geq M] = 0.5$

$\therefore P[\chi^2_{11} \geq M] = 0.5$ As shown in Fig. 2.4.

$P[\chi^2_{11} \geq 10.341] = 0.5$

$M = 10.341$

(v) Let D_2 and D_7 represent second and seventh deciles respectively then
 $P[Y \leq D_2] = 0.2$ $P[Y > D_2] = 0.8$

$P[Y > D_2] = 0.8$

$P[\chi^2 > 6.989] = 0.8$

$D_2 = 6.989$

As shown in Fig. 2.5,

Similarly, $P(Y \leq D_7) = 0.7$

$P(Y > D_7) = 0.3$

From Fig. 2.5, $P[\chi^2_{11} > 12.899] = 0.3$

$D_7 = 12.899$

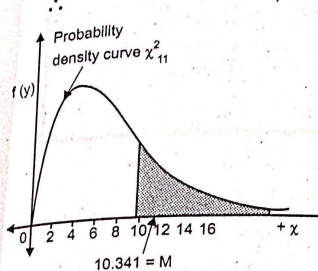


Fig. 2.4

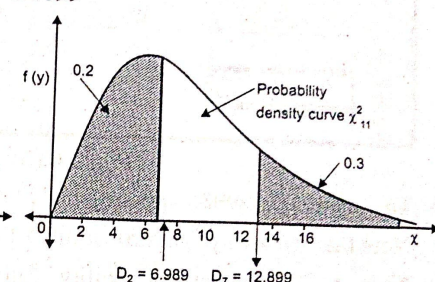


Fig. 2.5

(B) USE OF MS-EXCEL IN COMPUTING PROBABILITIES

Suppose we want to find $P[\chi^2_{11} > 8.148]$ using MS-EXCEL.

Step 1 : Click on **Insert** on the Excel sheet.

Step 2 : Click on **fx**.

Step 3 : Select function **CHIDIST**.

Then a box will appear on the screen as follows :

Function Arguments

CHIDIST

X: 8.148 = 8.148

Deg_freedom: 11 = 11

= 0.699988061

Returns the one-tailed probability of the chi-squared distribution.

Deg_freedom is the number of degrees of freedom, a number between 1 and 10^10, excluding 10^10.

Formula result = 0.699988061

Help on this function

OK Cancel

Fig. 2.6(a)

2.9 Probability Distribution of $\bar{X}$ and s^2

In majority cases a sample is assumed to be drawn from normal population. In order to develop tests regarding mean μ and variance σ^2 we need to find sampling distribution of mean $\bar{X}$ and sampling variance $S^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$. The derivation of distribution of S^2 needs orthogonal transformation and some results. Those are introduced and included in brief in the following discussion.

We study very interesting property that $\bar{X}$ and S^2 are independent random variables although they are algebraically dependent.

Transformation of Variables :

Suppose $X_1, X_2, \dots, X_n$ is a random sample, we define new set of variables $Y_1, Y_2, \dots, Y_n$ using linear transformation as follows :

[illegible]

Illustration 2 : If $X_1, X_2, \dots, X_{25}$ is a random sample from $N(5, 9)$, then find $P(3.8 \leq \bar{X} \leq 5.6, 5.637 \leq S^2 \leq 11.951)$.

Solution : Note that $\bar{X}$ and S^2 are independent random variables, hence

$$\begin{aligned} & P(3.8 \leq \bar{X} \leq 5.6, 5.637 \leq S^2 \leq 11.951) \\ &= P(3.8 \leq \bar{X} \leq 5.6) \cdot P(5.637 \leq S^2 \leq 11.951) \\ &= P\left(\frac{3.8-5}{3/5} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq \frac{5.6-5}{3/5}\right) \cdot P\left(\frac{25}{9} 5.637 \leq \frac{nS^2}{\sigma^2} \leq \frac{25}{9} \cdot 11.951\right) \\ &= P(-2 \leq N(0, 1) \leq 1) \cdot P(15.658 \leq \chi_{24}^2 \leq 33.197) \\ &= (1 - 0.15866 - 0.02275) \cdot [P(\chi_{24}^2 > 15.658) - P(\chi_{24}^2 > 33.197)] \\ &= 0.81859 \times (0.9 - 0.1) = 0.654872 \end{aligned}$$

Illustration 3 : Let X be a random sample of size 16 be drawn from a normal distribution with $\mu = 5$ and variance $\sigma^2 = 64$. Calculate $P(0 < \bar{X} < 10, 20.92 < S^2 < 122.32)$; $\bar{X}$ and S^2 being respectively the mean and variance of sample. (P.U. Oct. 2006)

Solution : Note that $\bar{X}$ and S^2 are independent random variables.

$$\text{Also } \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} = \frac{\bar{X}-5}{2} \rightarrow N(0, 1) \text{ and } \frac{nS^2}{\sigma^2} = \frac{16S^2}{64} = \frac{S^2}{4} \rightarrow \chi_{15}^2.$$

$$\begin{aligned} \therefore P(0 < \bar{X} < 10, 20.92 < S^2 < 122.32) \\ &= P\left[\frac{0-5}{2} < \frac{\bar{X}-5}{2} < \frac{10-5}{2}\right] P\left[\frac{20.92}{4} < \frac{S^2}{4} < \frac{122.32}{4}\right] \\ &= P[-2.5 < N(0, 1) < 2.5] P[5.23 < \chi_{15}^2 < 30.58] \\ &= [1 - P(N(0, 1) > 2.5)] P[\chi_{15}^2 \geq 5.23] - P[\chi_{15}^2 \geq 30.58] \\ &= [1 - 2(0.0062097)] [0.99 - 0.01] \\ &= (0.9875806)(0.98) = 0.967829 \end{aligned}$$

Illustration 4 : Let $\bar{X}$ and S^2 be the mean and variance of a random sample of size 25 from $N(3, 100)$ distribution.

Evaluate $P[0 < \bar{X} < 6, 55.2 < S^2 < 145.6]$

Solution : $P[0 < \bar{X} < 6, 55.2 < S^2 < 145.6]$

$$= P[0 < \bar{X} < 6] P[55.2 < S^2 < 145.6] \quad (\because \bar{X} \text{ and } S^2 \text{ are independent})$$

$$= P\left[\frac{0-3}{2} < \frac{\bar{X}-3}{2} < \frac{6-3}{2}\right] P\left[\frac{55.2}{4} < \frac{S^2}{4} < \frac{145.6}{4}\right]$$

$$= P\left[-\frac{3}{2} < N(0, 1) < \frac{3}{2}\right] P[13.8 < \chi_{24}^2 < 36.4]$$

$$= [1 - 2(0.06807)] [P(\chi_{24}^2 \geq 13.8) - P(\chi_{24}^2 \geq 36.4)]$$

$$= [0.863859] [0.99 - 0.05] = 0.812027$$

$$\text{Note : } \frac{\bar{X}-3}{10/5} = \frac{\bar{X}-3}{2} \rightarrow N(0, 1) \text{ and } \frac{nS^2}{\sigma^2} = \frac{25S^2}{100} = \frac{S^2}{4}$$

Solved Examples

Example 2.1 : If $Y \rightarrow \chi_{16}^2$, find

(i) $P[Y \leq 20.465]$ (ii) $P[11.152 \leq Y \leq 26.296]$ (iii) median of Y .

Solution :

$$\begin{aligned} \text{(i) } P[Y \leq 20.465] &= 1 - P[Y > 20.465] \\ &= 1 - P[\chi_{16}^2 > 20.465] = 1 - 0.2 \\ &= 0.8 \end{aligned}$$

$$\begin{aligned} \text{(ii) } P[11.152 \leq Y \leq 26.296] &= P[Y \geq 11.152] - P[Y \geq 26.296] \\ &= 0.8 - 0.05 = 0.75 \end{aligned}$$

(iii) Suppose median is M . Then $P[Y \geq M] = 0.8$

$$\therefore P[\chi_{16}^2 \geq M] = 0.5$$

$$\therefore M = 15.338$$

Example 2.2 : If $X_1, X_2, \dots, X_{113}$ are i.i.d. standard normal variates calculate :

$$P\left[\sum_{i=1}^{113} X_i^2 \leq 128\right]$$

$$Y = \sum_{i=1}^{113} X_i^2 \rightarrow \chi_{113}^2$$

$$\therefore P\left[\sum_{i=1}^{113} X_i^2 \leq 128\right] = P[\chi_{113}^2 \leq 128]$$

Since table values for χ_{113}^2 are not available, we use normal approximation.

$$\begin{aligned} &= P[\sqrt{2Y} \leq \sqrt{256}] \text{ By Fisher's approximation} \\ &= P[\sqrt{2Y} - \sqrt{2n-1} \leq \sqrt{256} - \sqrt{226-1}] \\ &= P[N(0, 1) \leq 16 - 15] \\ &= P[N(0, 1) \leq 1] \\ &= 1 - 0.15866 = 0.84134 \end{aligned}$$

Example 2.3 : Let $X_1, X_2, \dots, X_{20}$ be a random sample from normal population with mean $\mu = 5$ and variance $\sigma^2 = 16$

$$\text{Determine } P\left[\sum_{i=1}^{20} (X_i - 5)^2 \geq 364.4\right]$$

$$\text{Solution : Let, } Y = \sum_{i=1}^{20} \left(\frac{X_i - 5}{4}\right)^2 = \frac{\sum_{i=1}^{20} (X_i - 5)^2}{16}$$

$$\therefore X_i \rightarrow N(\mu = 5, \sigma^2 = 4^2), i = 1 \dots 20$$

$$Y \rightarrow \chi_{20}^2$$

$$\begin{aligned} \therefore P\left[\sum_{i=1}^{20} (X_i - 5)^2 \geq 364.4\right] &= P\left[\frac{\sum_{i=1}^{20} (X_i - 5)^2}{16} \geq \frac{364.4}{16}\right] \\ &= P[\chi_{20}^2 \geq 22.775] \\ &= 0.30 \end{aligned}$$

Example 2.4 : Let $X_1, X_2, \dots, X_i, \dots, X_{10}$ be independent normal variates such that $E(X_i) = 0$ and $\text{Var}(X_i) = i^2$.

$$\text{Find } P\left[X_1^2 + \frac{X_2^2}{2^2} + \frac{X_3^2}{3^2} + \dots + \frac{X_{10}^2}{10^2} \geq 7.267\right]$$

$$\text{Solution : } \because X_i \rightarrow N(0, i^2), \frac{X_i^2}{i^2} \rightarrow \chi_1^2$$

$$\therefore \sum_{i=1}^{10} \frac{X_i^2}{i^2} \rightarrow \chi_{10}^2$$

$$\begin{aligned} \therefore P\left[X_1^2 + \frac{X_2^2}{2^2} + \frac{X_3^2}{3^2} + \dots + \frac{X_{10}^2}{10^2} \geq 7.267\right] &= P\left[\sum_{i=1}^{10} \frac{X_i^2}{i^2} \geq 7.267\right] \\ &= P[\chi_{10}^2 \geq 7.267] \\ &= 0.7 \end{aligned}$$

Example 2.5 : If $X_1 \rightarrow N(4, 1)$, $X_2 \rightarrow N(4, 1)$ and X_1, X_2 are independent, find

$$P[0.296 < (X_1 - X_2)^2 < 0.910]$$

(P.U. April 2000)

Solution : $X_1 - X_2 \rightarrow N(0, 2)$

$$Y = \frac{(X_1 - X_2)^2}{2} \rightarrow \chi_1^2$$

$$\begin{aligned} P[0.296 < (X_1 - X_2)^2 < 0.910] &= P\left[\frac{0.296}{2} < \frac{(X_1 - X_2)^2}{2} < \frac{0.910}{2}\right] \\ &= P[0.148 < \chi_1^2 < 0.455] \\ &= P[\chi_1^2 \geq 0.148] - P[\chi_1^2 \geq 0.455] \\ &= 0.7 - 0.5 \\ &= 0.2 \end{aligned}$$

Example 2.6 : If $X \rightarrow \chi_2^2$, $Y \rightarrow \chi_2^2$ and X, Y are independent then find

$$(i) P\left(X + Y \leq 1.064, \frac{X}{Y} > 1\right)$$

$$(ii) \left[X + Y \leq 2.195, \frac{X}{X + Y} < \frac{1}{2}\right]$$

Solution : (i) Let, $U = X + Y$ $Y = \frac{X}{V}$ By result (2), U and V are independent. Also $U \rightarrow \chi_4^2$ and V has p.d.f. given by,

$$f(v) = \frac{1}{B(1, 1)} \cdot \frac{1}{(1+v)^2} \quad ; \quad 0 < v < \infty$$

$$\therefore P\left(X + Y \leq 1.064, \frac{X}{Y} > 1\right) = P(U \leq 1.064, V > 1)$$

Points to Remember

1. The p.d.f. chi-square distribution with n d.f. is

$$f(y) = \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma\left(\frac{n}{2}\right)} e^{-y/2} y^{\frac{n}{2}-1} ; y \geq 0.$$

2. If $y \rightarrow \chi_n^2$ then

(a) First 4 raw moments are given by

$$\mu_1' = \text{mean} = n, \quad \mu_2' = n(n+1), \quad \mu_3' = n(n+2)(n+4), \\ \mu_4' = n(n+2)(n+4)(n+6).$$

(b) First 4 central moments are given by

$$\mu_1 = 0, \quad \mu_2 = 2n, \quad \mu_3 = 8n, \quad \mu_4 = 48n + 12n^2.$$

$$(c) \beta_1 = \frac{8}{n}, \quad \gamma_1 = \sqrt{\frac{8}{n}}, \quad \beta_2 = 3 + \frac{12}{n}, \quad \gamma_2 = \frac{12}{n}.$$

(d) The r^{th} raw moments is

$$\mu_r' = n(n+2) \dots (n+2r-2) ; r = 1, 2 \dots$$

(e) The moment generating function (m.g.f.) is given by

$$M_X(t) = (1-2t)^{-n/2} ; t < \frac{1}{2}$$

(f) The cumulant generating function (c.g.f.) is

$$K_X(t) = -\frac{n}{2} \ln(1-2t) \text{ and } r^{\text{th}} \text{ cumulant}$$

$$K_r = (r-1)! n 2^{r-1} ; r = 1, 2 \dots$$

(g) The recurrence relation among the central moments is

$$\mu_{r+1} = 2r [n \mu_{r-1} + \mu_r] ; r = 1, 2, 3 \dots$$

(h) The mode of the distribution is $n - 2$ for $n > 2$.

$$(i) \frac{Y - E(Y)}{\sqrt{\text{Var}(Y)}} = \frac{Y - n}{\sqrt{2n}} \rightarrow N(0, 1) \text{ as } n \rightarrow \infty \text{ [normal approximation]}$$

(j) $\sqrt{2Y} - \sqrt{2n - 1} \rightarrow N(0, 1)$ for large n [Fisher's approximation]

3. **Additive property :** If Y_1 and Y_2 are independent χ^2 variates with n_1 and n_2 d.f. respectively then $Y_1 + Y_2$ has also χ^2 distribution with $(n_1 + n_2)$ degrees of freedom. The generalization of this property also holds good.

4. Important Results :

(i) If X and Y are independent chi-square variates with m and n d.f. respectively the $V = \frac{X}{X + Y}$ and $U = X + Y$ are independently distributed. Similarly $U = X + Y$ and $V = \frac{Y}{X + Y}$ are independently distributed. In this case $U \rightarrow \chi_{m+n}^2$.

(ii) If X and Y are independent variates with m and n d.f. respectively then $U = X + Y$ and $V = \frac{X}{Y}$ are independently distributed. Similarly $U = X + Y$ and $V = \frac{Y}{X}$ are also independently distributed. Here $U = \chi_{m+n}^2$.

(iii) If X is a continuous random variable with probability density function

$$f(x) = \frac{1}{B(m, n)} X^{m-1} (1 - X)^{n-1}; 0 \leq x \leq 1 \text{ with } m > 0, n > 0$$

$$= 0 \quad ; \text{ otherwise}$$

then for large m and comparatively small n , $-2m \log_e X$ follows approximately χ^2 distribution with $2n$ d.f.

(iv) Let X_1 and X_2 be two independent $N(0,1)$ variates. Then $Y_1 = \left(\frac{X_1 + X_2}{2}\right)^2$ and $Y_2 = \left(\frac{X_1 - X_2}{2}\right)^2$ are independent Chi-square variates with 1 d.f.

If $X_1, X_2, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$ then

(i) Sample mean $\bar{X}$ and sample variance $s^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ are independently distributed.

(ii) $\bar{X}$ follows $N\left(\mu, \frac{\sigma^2}{n}\right)$.

(iii) $\frac{ns^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}$ Follows χ^2 distribution with $(n - 1)$ d.f.